

Dingyun Huang, PhD

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EDUCATION

- 2021 – 2026 **University of Cambridge – PhD in Physics**
Data-driven methods for applications in optoelectronics
- 2017 – 2021 **University of Cambridge – BA & MSci in Natural Sciences**
1st class honours with specializations in Physics
- 2015 – 2017 **Lime House School – A Levels**

TECHNICAL SKILLS

- ▶ **Linux & Supercomputers:** command line tools, shell scripts, exascale supercomputing clusters; MPI-based multi-process workflows; multi-GPU and multi-node distributed model training; website hosting through AWS.
- ▶ **Python:** extensively used PyTorch, Scikit-Learn, Pandas, and other popular data-science/ML packages; large-scale data manipulation and visualization; experienced in object-oriented programming and code packaging for managing ChemDataExtractor and group websites.
- ▶ **ML Algorithms:** CNN, GNN, Transformers, LSTM, Random Forests, Bayesian methods, feature engineering.
- ▶ **Language Modelling:** conventional NLP methods; pre-training and fine-tuning transformer-based language models, such as BERT and Llama; deploying language models on downstream applications such as knowledge extraction through multi-turn QA and RAG.
- ▶ **Material modelling:** cheminformatics toolkit RDKit; Cambridge Structural Database; property modelling using graph neural networks and other ML algorithms.
- ▶ **Git & PyPI:** version management and code collaboration on group software through GitHub; feature unit testing and package publishing via PyPI.
- ▶ **Mathematics:** postgraduate level calculus, tensor algebra, statistics, differential equations, probabilities
- ▶ **Language:** Chinese (native); English (fluent); Cantonese (basic)

RESEARCH PROJECTS

University of Cambridge – Research Associate

- 2025-2026 **Optimization of galvanizing line control with deep learning, collaboration with Tata Steel UK**
- Devised data pipeline to align, clean, and resample production line signals and measurements
 - Trained a pair of prediction and feedback models that optimize production parameters according to the target and achieved coating weights on-the-fly.

University of Cambridge – PhD in Physics

- 2024 – 2025 **End-to-end workflow of training machine learning predictors for PL wavelengths**
- Extracted a dataset of experimental PL wavelengths from the literature at **above 90% precision**.
 - Developed a solution for extracting data from Wiley articles PDFs, leading to **20% more data**.
 - Designed and trained a GNN model to predict PL wavelengths of TADF molecules at **0.12 eV MAE**.
 - Published results in *ACS J. Chem. Inf. Model.* doi/10.1021/acs.jcim.6c00425
- 2023 – 2024 **Large language models for optoelectronics**
- Pre-trained and finetuned three BERT-like language models for optoelectronics tasks, including data-mining through multi-turn Q&A, and document retrieval.
 - Conducted a detailed comparison of the efficiency of domain adaptation technique for BERT and RoBERTa; Showed that models adapted from RoBERTa required an **80% shorter** pre-training than those from BERT to reach similar performance.
 - Published results in *ACS J. Chem. Inf. Model.* doi.org/10.1021/acs.jcim.4c02029
- 2021 – 2023 **Automatic database collection for TADF materials through text-mining**
- Developed an automated **text-mining pipeline** basing on ChemDataExtractor, achieving **82% precision** in curating an OLED materials database. The database has been downloaded over 500 times since its release in 2024.

- Innovated a statistical screening method for compound name extraction, boosting the corresponding precision **from 30% to 80%**.
 - Designed an OPSIN-based workflow of 92% accuracy on deducing molecular structure of a compound from journal texts; this package PyOPSIN now has a **thousand downloads** on PyPI.
 - Published results in *Nature Scientific Data* doi.org/10.1038/s41597-023-02897-3 **cited by 12**
- 2021 - present **ChemDataExtractor**
- Located and fixed multiple critical issues in ChemDataExtractor modules of abbreviation detections, text and table parsing, and XML scraping, boosting extraction recall **by 30%**.
 - Actively maintained the group software ChemDataExtractor2, resolving issues raised on GitHub.
 - Successfully migrated ChemDataExtractor dependencies from AllenNLP to Torch and HuggingFace implementations and upgraded the software from Python 3.6 to the active Python 3.11.

EXPERIENCES

- 2025 – 2026 **Department of Physics, University of Cambridge – Research Associate**
- Developed a **time-series** model to accurately predict coating weight from production parameters.
 - Proposed and trained a counter-part model that updates parameters when coating weight drifts.
- Summer 2025 **Huawei Hong Kong Research Centre, Theory Lab – Research Scientist Internship**
- Developed physics-informed deep learning methods for optical imaging systems optimizations.
 - Reduced the regression error by **65%** by a new model architecture and training setup while reducing the number of trainable parameters by **90%** (Details are confidential subject to NDA).
- 2024 – 2025 **Churchill College, University of Cambridge – Mathematics Supervisor**
- Guided six first-year Natural Sciences students in Mathematics B, simplifying advanced concepts and improving problem-solving skills. Received positive feedback from students for fostering a supportive and effective learning environment.
- Apr – Sep 2022 **University of Cambridge STEM Smart Project – Supervisor**
- Designed and delivered interactive pre-university mathematics lessons for non-fee-paying six form students, contributing to the Cambridge STEM Smart initiative. Helped bridge knowledge gaps for underrepresented students, boosting their preparedness for subject challenges and higher education.
- Jan – Mar 2022 **University of Cambridge – Python Computing Demonstrator**
- Introduced numerical methods used in physics and their real-world applications. Developed tailored exercises to address varying skill levels, resulting in a marked improvement in students' programming proficiency.

INTERESTS & ACTIVITIES

- 2021 – present Churchill College Basketball Team (topped the college league for 5 seasons)
- 2019 and 2022 Oriental Dance Association Treasurer (2022), Publicity and Sponsorship Officer (2019)
- 2018 – 2019 Cambridge University Chinese Cultural Society Sports Officer

SCHOLARSHIPS & AWARDS

- 2025 **Royal Society of Chemistry Materials Chemistry Horizon Prize** as a part of the team led by Prof. Jacqueline Cole for the development of chemistry-aware artificial intelligence software, its application to data-driven materials discovery, and its provision of open-source materials databases and language models for the global scientific community.
- 2021 – 2025 **Cambridge Trust Scholarship** towards a PhD at the University of Cambridge
- 2017 – 2021 **Churchill College Scholarship** for outstanding performance in Tripos examinations
- 2016 – 2017 **Gold Awards** (UK top **50**) in the British Physics and Astronomy & Astrophysics Olympiads